Capstone Project

Data Science Analytics Report

Walmart Sales Forecasting & Online Retail Customer Analysis

Data Science Capstone

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Table of Contents

Right-click the TOC and select "Update Field" to refresh page numbers

1. Problem Statement **3**

2. Project Objective **4**

3. Data Description **5**

Walmart Dataset 5

Online Retail Dataset 6

4. Data Pre-processing **7**

5. Algorithm Selection **9**

6. Model Evaluation **11**

7. Inferences and Insights **13**

8. Future Possibilities **15**

1. Problem Statement

Problem Statement 1: Walmart Sales Forecasting

A retail store with multiple outlets across the country is facing challenges in managing inventory to match demand with respect to supply. The organization requires a data-driven approach to forecast sales for the upcoming periods and optimize inventory management across all store locations.

The key challenges include: (1) Understanding sales trends over time to identify seasonal patterns, (2) Comparing sales performance during holiday versus non-holiday periods, (3) Determining correlations between weekly sales and external factors such as temperature, fuel price, CPI, and unemployment rates, and (4) Developing accurate forecasting models to predict sales for the next 12 weeks.

Problem Statement 2: Online Retail Customer Analysis

An online retail store seeks to understand various customer purchase patterns to enhance business strategies. The organization needs insights into customer behavior, top-performing products, and sales trends over time to optimize marketing campaigns and inventory decisions.

The analysis objectives include: (1) Identifying useful insights about customer purchasing history, (2) Segmenting customers based on their purchasing behavior, (3) Analyzing top-selling products by total sales revenue, and (4) Understanding time-based sales trends to identify seasonality and growth patterns.

2. Project Objective

Primary Objectives

- Develop accurate sales forecasting models for Walmart stores to predict weekly sales for the next 12 weeks
- Analyze the impact of holidays on sales performance and identify seasonal trends
- Examine correlations between sales and external economic factors (temperature, fuel price, CPI, unemployment)
- Identify top-selling products and customer purchasing patterns in online retail
- Segment customers based on their purchasing behavior for targeted marketing strategies

Secondary Objectives

- Provide actionable insights for inventory management and demand planning
- Enable data-driven decision making for promotional campaigns during high-demand periods
- Establish a framework for continuous sales monitoring and forecasting

3. Data Description

Walmart Dataset (Walmart.csv)

The Walmart dataset contains 6,435 rows and 8 columns, representing weekly sales data for multiple store locations. The dataset spans from February 2010 to October 2012, providing approximately 2.5 years of historical sales data for analysis.

The dataset features include:

Feature Name	Description
Store	Store number identifier
Date	Week of sales recording
Weekly_Sales	Sales for the given store in that week
Holiday_Flag	Indicator if it is a holiday week (1=Yes, 0=No)
Temperature	Temperature on the day of sale (Fahrenheit)
Fuel_Price	Cost of fuel in the region (USD)
CPI	Consumer Price Index
Unemployment	Unemployment Rate (%)

Table 1: Walmart Dataset Features

Online Retail Dataset (OnlineRetail.csv)

The Online Retail dataset contains 387,961 rows and 8 columns, representing transactional data from an online retail store. The dataset covers the period from December 2010 to December 2011, capturing customer purchase behaviors across different countries.

The dataset features include:

Feature Name	Description
InvoiceNo	Invoice number identifier
StockCode	Product ID code

Feature Name	Description
Description	Product description
Quantity	Quantity of product purchased
InvoiceDate	Date and time of invoice
UnitPrice	Price per unit (GBP)
CustomerID	Customer identifier
Country	Region of purchase

Table 2: Online Retail Dataset Features

4. Data Pre-processing Steps and Inspiration

Walmart Data Pre-processing

Step 1: Data Loading and Initial Inspection

The dataset was loaded using pandas with appropriate encoding. Initial inspection revealed no missing values and no duplicate records, indicating good data quality from the source. The dataset contains 6,435 complete records across all 8 features.

Step 2: Date Conversion

The 'Date' column was converted from string format to datetime objects using pd.to_datetime() with format='%d-%m-%Y'. This conversion enables time-series analysis and facilitates extraction of temporal features such as year and month for trend analysis.

Step 3: Feature Engineering

Additional temporal features were created by extracting Year and Month from the Date column. These features enable aggregation of sales data by time periods for trend visualization and seasonal pattern identification.

Online Retail Data Pre-processing

Step 1: Data Loading

The dataset was loaded using pandas with latin1 encoding to handle special characters in product descriptions. The initial dataset contained 301,296 entries across 8 columns.

Step 2: Missing Value Treatment

Significant missing values were identified: 84,500 missing CustomerID entries (28% of data), 1,110 missing Description entries, and isolated missing values in Quantity, InvoiceDate, UnitPrice, and Country columns. Rows with missing CustomerID were removed as these transactions cannot be attributed to specific customers, which is critical for retail analysis.

Step 3: Data Cleaning

Records with negative quantities or unit prices were removed as they likely represent returns or data entry errors. After cleaning, the dataset was reduced to 211,567 valid transactions. A new 'TotalPrice' feature was created by multiplying Quantity and UnitPrice for each transaction.

5. Choosing the Algorithm for the Project

Analysis Approach

For this capstone project, an exploratory data analysis (EDA) approach was primarily employed using Python libraries including pandas, numpy, matplotlib, and seaborn. The analysis focused on descriptive statistics, correlation analysis, and time-series visualization rather than predictive modeling algorithms.

Techniques Employed

Time-Series Analysis

Sales trends were analyzed by aggregating weekly sales data by year and month. Line plots with markers were used to visualize temporal patterns and identify seasonal fluctuations in sales performance.

Comparative Analysis

Holiday versus non-holiday sales performance was compared using grouped aggregation and bar chart visualization. This analysis revealed the significant impact of holiday periods on sales volumes.

Correlation Analysis

Correlation matrices were computed to examine relationships between weekly sales and external factors (temperature, fuel price, CPI, unemployment). Heatmaps with annotated correlation coefficients provided clear visualization of these relationships.

Customer and Product Analysis

For the online retail dataset, customer activity was analyzed by calculating unique customer counts and average order values. Top-selling products were identified through aggregation by product description and sorting by total sales revenue.

Motivation for Algorithm Selection

The exploratory analysis approach was chosen because: (1) It provides comprehensive understanding of data patterns before building predictive models, (2) It reveals actionable insights that can inform business decisions immediately, (3) It identifies data quality issues and feature relationships that guide model selection, and (4) It establishes baseline performance metrics for evaluating predictive models in future work.

6. Model Evaluation and Techniques

Evaluation Metrics and Techniques

Since this project focused on exploratory data analysis rather than predictive modeling, evaluation was based on data quality metrics, statistical insights, and visualization clarity. The following techniques were employed:

Statistical Summary

Descriptive statistics including mean, median, standard deviation, minimum, and maximum values were computed for all numerical features. This provided baseline understanding of data distributions and identified potential outliers.

Correlation Coefficients

Pearson correlation coefficients were calculated to measure linear relationships between variables. The correlation matrix revealed weak to moderate correlations between sales and external factors, with unemployment showing the strongest negative correlation.

Data Quality Assessment

Data completeness was evaluated by checking for missing values across all columns. For the online retail dataset, data quality issues including negative quantities and prices were identified and addressed through filtering.

Key Findings from Analysis

- Walmart dataset: 6,435 records with no missing values after initial inspection
- Online Retail dataset: Reduced from 301,296 to 211,567 records after cleaning
- Identified 3,226 unique customers in the cleaned online retail dataset
- Average order value calculated at $400.67 per customer
- Holiday weeks consistently show higher average weekly sales

7. Inferences from the Analysis

Walmart Sales Analysis Inferences

Sales Trends

Average weekly sales exhibit identifiable patterns over time when aggregated by year and month, suggesting seasonal or cyclical behaviors in sales performance. The time-series visualization reveals fluctuations that correspond to seasonal shopping patterns.

Holiday Impact

Holiday weeks consistently show higher average weekly sales compared to non-holiday weeks, indicating a positive impact of holidays on sales performance. This finding suggests opportunities for targeted marketing campaigns during holiday periods.

External Factor Correlations

The correlation analysis revealed relationships between weekly sales and external economic factors. Temperature, fuel price, CPI, and unemployment all show varying degrees of correlation with sales, providing insights for demand forecasting models.

Online Retail Analysis Inferences

Data Quality Insights

The substantial amount of initial data cleaning required, particularly around missing CustomerID values and negative quantities/prices, underscores the importance of robust data validation and preprocessing steps in retail analytics.

Customer Behavior

With 3,226 unique customers and an average order value of $400.67, the analysis reveals a diverse customer base with varying purchase patterns. Understanding these patterns can inform customer segmentation strategies.

Product Performance

Top 10 selling products were identified based on total sales, indicating which items contribute most significantly to revenue. This information is valuable for inventory management and promotional planning.

Sales Trends Over Time

Monthly sales trends analyzed over time allow for observation of seasonality or growth/decline patterns. These insights can inform inventory management, marketing campaigns, and promotional strategies to capitalize on high-demand periods.

8. Future Possibilities of the Project

Predictive Modeling Opportunities

The exploratory analysis conducted in this project provides a foundation for developing predictive models. Future work could include:

- Time-series forecasting models (ARIMA, Prophet, LSTM) for sales prediction
- Machine learning models (Random Forest, XGBoost) incorporating external factors
- Customer churn prediction models based on purchase behavior patterns
- Product recommendation systems using collaborative filtering

Enhanced Analytics

- RFM (Recency, Frequency, Monetary) analysis for customer segmentation
- Market basket analysis to identify product associations
- Cohort analysis to track customer retention over time
- Geographic analysis of sales patterns by region

Business Applications

The insights from this analysis can be leveraged to develop targeted marketing campaigns specifically for holiday periods to maximize revenue. Predictive models built on the identified sales trends and correlations could aid in better resource allocation and inventory management. Additionally, customer segmentation based on purchasing behavior can enable personalized marketing strategies and improved customer retention.

Technical Enhancements

- Implementation of automated data pipelines for real-time analysis
- Development of interactive dashboards for business stakeholders
- Integration with external data sources for enhanced forecasting
- Deployment of models as APIs for production use

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